SCHOOL OF ENGINEERING

Dayananda Sagar University Bengaluru

RED TAPE
Automated Bureaucracy & Contract Execution Engine
SECTION 6D

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REDTAPE

- NLP-powered document automation
- Contract risk detection
- Pre-litigation workflow engine
- Backend-first scalable architecture

Problem Statement

- Legal processes are confusing
- Grievances remain unstructured
- Contracts contain hidden risk clauses
- Users lack procedural clarity
- No execution-focused tools

Gap: Advice exists. Execution does not.

Solution Architecture

System Overview

User Story → NLP Extraction → Structured Schema → Legal Notice PDF → Action Tracker

Core Modules:

- Story-to-Form NLP Pipeline
- Contract Red Flag Scanner
- Dispute Classification Engine
- Escalation Workflow Tracker

Backend: FastAPI + spaCy + ReportLab
Deployment: Cloud API + Mobile Client

Story to Form Pipeline

NLP EXECUTION ENGINE

- Unstructured input ingestion
- Named Entity Recognition (NER)
- Custom entity mapping
- Pydantic schema validation
- Automated PDF generation

Story to Form Pipeline

OUTPUT

- Legally formatted draft notice
- Structured dispute metadata
- Evidence-ready documentation

RISK DETECTION SYSTEM

- Clause extraction
- Text classification
- Predatory term identification
- Plain-English risk explanation

USE CASES

- Internship agreements
- Rental contracts
- NDAs
- Employment offers

Goal: Reduce asymmetric information

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ENGINEERING DESIGN

- Modular service architecture
- Backend-first deployment model
- Scalable API design
- Clean separation of concerns
- Production-ready ML integration

IMPACT

- Reduces procedural friction
- Converts confusion into action
- Bridges law and automation

Thank You